

A Two-Dimensional Counterexample for the SEP and Beta TTRS Relaxations

Computational note

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This note records the small counterexample referenced at the end of Section 4.2 of the current paper draft. It is written in the notation of the paper: $x \in \mathbb{R}^n$ is the original TTRS variable, $A = \text{Diag}(a)$ is diagonal positive definite, and the second ball constraint is written through $y = Ax + b$.

1 Instance

Consider

$$\begin{aligned} \min \quad & x^T Q x + c^T x \\ \text{s.t.} \quad & \|x\| \leq 1, \\ & \|Ax + b\| \leq 1, \end{aligned} \tag{1}$$

where $n = 2$,

$$a = \begin{bmatrix} 4/3 \\ 13/12 \end{bmatrix}, \quad b = \begin{bmatrix} 11/20 \\ -1/20 \end{bmatrix},$$

and

$$Q = \begin{bmatrix} -24/16 & -1/16 \\ -1/16 & -13/16 \end{bmatrix}, \quad c = \begin{bmatrix} -1 \\ 0 \end{bmatrix}.$$

The smallest eigenvalue of Q is approximately -1.505635621484 , so the objective is nonconvex.

2 SEP Relaxation

The Full SEP relaxation uses the Shor moment matrix

$$H = \begin{bmatrix} 1 & x^T \\ x & X \end{bmatrix} \succeq 0$$

together with the two trace inequalities

$$\text{tr}(X) \leq 1, \quad \text{tr}(A^2 X) + 2b^T A x + \|b\|^2 \leq 1.$$

It also introduces

$$Z = \begin{bmatrix} 1 & x^T \\ y & V \end{bmatrix} \in \text{SEP}(n+1, n+1),$$

where $y = Ax + b$, and imposes the affine consistency equations

$$Z_{0j} = x_j, \quad Z_{i0} = a_i x_i + b_i, \quad Z_{ij} = a_i X_{ij} + b_i x_j.$$

The objective is $\langle Q, X \rangle + c^T x$. On this instance, the Full SEP relaxation gives

$$z_{\text{SEP}} = -0.632844508289.$$

3 Shifted Beta Relaxation

The shifted beta relaxation introduces variables β_i representing x_i^2 . With $w = (1, x, \beta)$, the two ball constraints become

$$1 - \sum_i \beta_i \geq 0,$$

and

$$1 - \|b\|^2 - 2 \sum_i a_i b_i x_i - \sum_i a_i^2 \beta_i \geq 0.$$

The inequalities $\beta_i \geq x_i^2$ are imposed with rotated second-order cones, and the beta SDP then includes the Shor, RLT, SOC-RLT, and Kronecker constraints in the same spirit as the beta relaxation used for the concentric TTRS instances. For this shifted instance, the relaxation gives

$$z_\beta = -0.550899217259.$$

4 True Optimum

The true optimum was certified by enumerating the one-dimensional boundary arcs of the feasible intersection. The value is

$$z^\star = -0.546797627007,$$

attained at approximately

$$x^\star = \begin{bmatrix} 0.185929524305185 \\ 0.602567865182517 \end{bmatrix}.$$

Thus both relaxations are strict lower bounds:

$$z^\star - z_{\text{SEP}} = 0.086046881282, \quad z^\star - z_\beta = 0.004101590252.$$

5 Diagnostic Values

At the Full SEP optimum, the first-column point is well inside both ball constraints:

$$\|x\|^2 = 0.139541363013, \quad \|Ax + b\|^2 = 0.004339906310.$$

The trace quantities behave differently:

$$\text{tr}(X) = 0.808104046269, \quad \text{tr}(A^2 X) + 2b^T A x + \|b\|^2 = 1.000000000282.$$

The rank-one residual of the Shor moment matrix is approximately

$$0.498013951456.$$

This is consistent with the interpretation that the missing ingredient is synchronization between the lifted quadratic representation and the product cone representation.

6 Reproduction

The script

`paper-code/verify_ttrs_sep_beta_counterexample.py`

re-solves the Shor relaxation, the Full SEP relaxation, and the shifted beta relaxation, and then certifies the true value by boundary enumeration. It is a named reproducibility check rather than part of the random computational panels.